RiskPredictor: Pricing and Risk for Sports Prediction Markets

Research report

Sai Kishore Pachigolla
NYU Tandon, Finance and Risk Engineering
Advisor: Prof. Dennis Shasha, Courant Institute

Last updated: 28 October 2026

Abstract

Prediction markets on sporting events now trade billions of contracts a season, yet the risk tools of traditional finance assume continuous prices and long histories that binary, short-lived contracts do not have. This report records a project to build a risk engine for such markets. Its central idea is that a contract's price is a probability, so pricing and settlement risk are the same calculation, and a joint model over a game's final score prices every contract on the game consistently and lets risk net across them. So far we have classified which traditional risk tools carry over; surveyed the literature on predicting NFL games; built a leak-free table of 759 pre-game features for 7,548 games (1999–2026); and fitted a two-output random forest that predicts both teams' scores. Evaluated walk-forward over sixteen seasons it predicts the margin with a mean absolute error of 10.35 points, against 10.38 for Elo and 10.05 for the Vegas closing line, matching the published literature. Its score predictions are calibrated, its residuals are uncorrelated with standard deviations of 9.6 and 9.4 points, and, placed on an empirical score grid that preserves the NFL's key-number spikes, they price a game's full ladder of contracts to within a cent of the Kalshi order book. Ensembles with public forecasts do not beat the closing line. The engine's value therefore lies in a calibrated joint distribution rather than a better point forecast. The report is updated weekly; Section 3 holds the dated progress reports.

Contents

1 The idea **2**

1.1 A contract's price is a probability, so pricing and risk are one model 2

1.2 VaR and expected shortfall for binary contracts 4

1.3 Which traditional risk tools carry over 4

2 Data **6**

2.1 Kalshi 6

2.2 NFL results and the pre-game feature table 7

2.3 Testing 8

3 Weekly progress **8**

3.1 Report of 30 September 2026: pricing as outcome modelling 8

3.2 Report of 7 October 2026: literature, feature table, two-output random forest 13
3.3 Report of 21 October 2026: ensembles and full distributions 20
3.4 Report of 28 October 2026: parlays 24

4 What we have learned, and next steps **29**
4.1 What we have learned 29
4.2 Next steps 30

1 The idea

Prediction markets have moved from a niche into the mainstream. On Kalshi alone the three main contract types on a single NFL week (game winner, point spread, game total) traded close to two billion contracts over the first three weeks of the 2026 season, and the Super Bowl winner market carries over a hundred million contracts of open interest. In this document we explore how the methods of traditional financial risk management apply to such markets and build a risk engine for them, *RiskPredictor*.

The goals of the engine are:

1. compute security-level and portfolio-level risk;
2. in computing portfolio risk, add or net exposures according to the defined behaviour of the assets (netting through correlation, or, as we argue below, through a joint outcome model);
3. produce metrics that show what a book would actually do when the market moves;
4. at a later stage, support margin, collateral and leverage calculations;
5. identify cross-market hedging opportunities that allow more balanced products and portfolios.

1.1 A contract's price is a probability, so pricing and risk are one model

A binary contract pays \$1 or nothing, so its fair value under any probability assessment π of the outcome is π dollars, and with only two outcomes that probability *is* the full profit-and-loss distribution (Section 1.2). Pricing and settlement risk are therefore the same calculation. For a portfolio, the pricing model must be a joint one, because the joint probabilities are what allow risk to net across positions.

Amine comment: Please distinguish the model's outcome probabilities from market prices. Treating them as equal requires assumptions beyond accounting for fees and spreads. I believe matching prices alone does not show that the probabilities accurately describe future outcomes.

Sai comment: Fixed. The paragraph below now separates the market price p from the model probability π, lists the assumptions under which they coincide, and states that outcome accuracy is judged only against results (Sections 3.2.2 and 3.3), with price matching read as consistency with the market, not correctness.

Two probabilities appear in this document and we keep them apart. The *market price* p of a contract is what the exchange's order book shows, net of fee and spread. The *model probability* π is our estimate of the chance the outcome occurs. The two coincide only under assumptions that go beyond fees: that the marginal trader is risk-neutral and uninformed traders' flow is absorbed without moving the price, and that the market is efficient in the sense that its price already uses all public information. Levitt [13] shows that sportsbooks deliberately set prices away from their

own probabilities to exploit bettor biases, so even an informed price need not equal its setter's probability. We therefore use market prices for two purposes only: as a benchmark that any model must be compared with, and as the mark at which a position is valued before settlement. Whether model probabilities describe future outcomes is a separate question, answered only by comparing them with what happened: the calibration plots and Brier scores of Sections 3.2.2 and 3.3 do that, and the market's own prices are held to the same test in Figure 1. Matching Kalshi's prices, as in the worked example of Section 3.1, shows that a model is consistent with the market, not that either is right.

A proper joint outcome model, such as a distribution over the final score of a game, gives everything in Table 1 at once.

Output	**Used for**
Fair price of every contract on the game	Pricing
Probability of each outcome state	ES, probability of loss, expected loss
Joint behaviour of contracts on the same game	Netting (goal 2 of the engine)
Consistent prices across winner, spread, total and props	Detecting mispricing between related contracts

Table 1: What a joint outcome model provides.

Every Kalshi contract on a game is a region of the final-score grid. With h and a the home and away scores, the winner contract pays when $h - a > 0$ (ties settle at \$0.50), a spread contract at strike k pays when $h - a > k$, and a total contract pays when $h + a > k$. One probability table $p(h, a)$ prices all of them by summing cells, and a book of positions is a payoff vector over the same grid.

Amine comment: Fitting spread and total prices does not establish how margin and total behave together.

Sai comment: Fixed. The paragraph below states that the ladders identify only the two marginals and that the dependence between margin and total is an assumption carried by the historical prior (parity, correlation 0.03, conditional widening), untestable against listed contracts.

The listed contracts do not, however, identify the whole table. Every winner and spread contract is a function of the margin $h - a$ alone and every total contract of $h + a$ alone, so a game's full ladder of prices pins down the two marginal distributions and says nothing about how margin and total move together. The dependence between them has to come from somewhere else. In our construction it comes from the historical prior: the empirical score table carries the parity constraint (margin and total are always both odd or both even), the near-zero linear correlation between them (0.03 over 2015–2026) and the modest widening of the margin in high-scoring games, and the tilt of Section 3.1 preserves all three because it only reweights cells. This is an assumption, stable across seasons in the data we have, not something the market confirms. It matters for any position whose payoff depends on both numbers at once, such as a spread and a total on the same game held together, and it could only be tested against contracts that pay on joint events (team totals, same-game parlays), which Kalshi does not list.

1.2 VaR and expected shortfall for binary contracts

For a binary contract none of the three traditional computations is needed, because the loss distribution can be written down. A position of q contracts bought at price p pays $q(1-p)$ if the event occurs and loses qp otherwise. We define the loss as minus the profit, $L = -\text{P\&L}$, so a gain is a negative loss; if the probability of the event is π, L takes the value qp with probability $1-\pi$ and $-q(1-p)$ with probability π. With $\text{VaR}_\alpha = \inf\{\ell : P(L \le \ell) \ge \alpha\}$ and $\text{ES}_\alpha = \frac{1}{1-\alpha}\int_\alpha^1 \text{VaR}_u \, du$,

$$\text{VaR}_\alpha = \begin{cases} qp & \text{if } \alpha > \pi, \\ -q(1-p) & \text{if } \alpha \le \pi, \end{cases} \qquad \text{ES}_\alpha = \begin{cases} qp & \text{if } \alpha \ge \pi, \\ \dfrac{(1-\pi)\,qp - (\pi-\alpha)\,q(1-p)}{1-\alpha} & \text{if } \alpha < \pi. \end{cases}$$

When the loss state is rarer than $1-\alpha$ (a likely event, $\pi > \alpha$), the α-quantile of the loss is the gain state and VaR is negative; the tail average then mixes the loss state with part of the gain state. An earlier draft truncated gains at zero, $L^+ = \max(L, 0)$, which leaves VaR at 0 rather than $-q(1-p)$ in that case and overstates ES by $(\pi-\alpha)\,q(1-p)/(1-\alpha)$; for the common case $\alpha \ge \pi$ the two conventions agree. So for a single contract the whole of risk measurement reduces to one number, π, which is also the fair price. For a portfolio of contracts on the same game the loss distribution is a sum over the joint outcomes of the game: with a probability table $p(h, a)$ over the final score, each position's payoff is a function of (h, a), the portfolio profit and loss is their sum cell by cell, and VaR, ES, the worst case and the probability of loss are read directly from the resulting distribution with no simulation and no covariance matrix. This is the sense in which the pricing model and the risk model are the same object, and it is why the rest of this document is about building $p(h, a)$. **Amine comment:** Please define loss explicitly. The formulas treat gains as zero; I believe defining loss as minus profit can give different VaR and expected shortfall values?

Sai comment: Fixed. Loss is now defined as $L = -\text{P\&L}$ and the VaR and ES formulas above are the signed-loss versions; the text notes where the earlier truncated-at-zero formulas differed (VaR of 0 instead of $-q(1-p)$, and ES overstated by $(\pi-\alpha)\,q(1-p)/(1-\alpha)$, when $\pi > \alpha$).

1.3 Which traditional risk tools carry over

Traditional risk tools were built for instruments whose prices move continuously and have long histories. Sports prediction contracts pay exactly \$1 or \$0, last hours or days, and move in jumps. As a result the *processes* of traditional risk management carry over, the *measures* need adapting, and the *statistical assumptions* no longer hold. Table 2 summarises our classification.

Category	Tools	Why
Works as-is	Expected shortfall, scenario revaluation, SPAN margining, stress testing, limit frameworks, counterparty exposure	They work for any payoff shape. Because every possible outcome can be listed, they can be computed exactly.
Needs adapting	Greeks, jump-to-default, basket correlation, netting, liquidity-adjusted risk, backtesting	The concepts are valid but must be reworked for binary, jump-driven payoffs, where dependence between contracts comes mostly from their structure rather than from statistics.
Does not work	VaR as usually computed, covariance matrices, factor models, historical price simulation, continuous hedging	They assume smooth, normally distributed returns with long price histories, none of which exist here.
New to prediction markets	Resolution-rule risk, cross-venue basis risk, integrity risk	These have no direct equivalent in traditional finance.

Table 2: Traditional risk tools and how they transfer to binary sports contracts.

1.3.1 Terms

Value at Risk (VaR) the loss that will not be exceeded at a given confidence level (e.g. 95%). It ignores how large losses can be beyond that point.
Expected shortfall (ES) the average loss in the worst outcomes beyond the VaR threshold, which captures the size of the tail.
Scenario revaluation repricing every position under each possible scenario and adding up the profit and loss.
SPAN margining setting collateral equal to the worst portfolio loss across a defined set of scenarios.
Stress testing measuring losses under extreme but plausible events. *Reverse* stress testing finds the events that would cause a given loss.
Greeks sensitivities of a position's value. *Delta* measures the response to price, *gamma* how delta itself changes, and *theta* the effect of time passing.
Basket correlation risk in a product that depends on several events together (e.g. a parlay), driven by how those events are related.
Netting offsetting gains and losses across related positions to find the true combined exposure.
Liquidity-adjusted risk risk measured at the price one could actually exit at, rather than the mid-price.
Covariance matrix, factor model statistical tools that estimate how assets move together from historical returns.
Counterparty exposure potential loss if a counterparty or venue fails to pay.
Resolution-rule risk loss from how a contract settles, e.g. voids, pushes, overtime or stat corrections.
Cross-venue basis risk the "same" contract settling differently on two platforms.
Integrity risk loss from insider information or match-fixing.

1.3.2 Value at Risk

VaR answers one question: how much could I lose over a given period, at a given confidence level, under normal market conditions? Every VaR number has three parts: a loss amount, a time horizon and a confidence level. It is traditionally computed in one of three ways: parametrically, assuming a normal distribution of the underlying; by historical simulation; or by Monte Carlo. Its limitations are well known. It says nothing about the size of losses beyond the threshold, so two portfolios can have the same VaR while one loses ten times more in the worst 1% of cases. It is not subadditive, so $\mathrm{VaR}(A+B)$ can exceed $\mathrm{VaR}(A) + \mathrm{VaR}(B)$, which makes diversification appear to increase risk; for this reason it is not a coherent risk measure in the sense of Artzner et al. [1]. And it relies on normal conditions, while correlations and volatilities shift sharply in a crisis.

2 Data

2.1 Kalshi

Kalshi is a regulated US exchange on which event contracts trade in a central limit order book, each contract paying $1 if the event occurs. Table 3 lists the NFL contract series with their volume at the time of our first survey. Game-level contracts (winner, spread, total) are the focus of this project because they are the ones a joint score model prices directly.

Series	Type	Markets	Volume	Open int.
KXSB	Super Bowl winner	32	131.9M	108.0M
KXNEXTTEAMNFL	Player next team	290	4.69M	2.86M
KXNFLGAME	Game winner (moneyline)	64	4.60M	3.93M
KXNFLDPOTY	Defensive player of the year	65	2.32M	2.13M
KXNFLPLAYOFF	Playoff qualifiers	32	1.85M	1.20M
KXNFLNFCEAST	Division winner	4	1.59M	1.17M
KXNFLSPREAD	Point spread	414	1.40M	1.22M
KXNFLTOTAL	Game total	304	1.16M	1.12M
KXNFLAFCNORTH	Division winner	4	0.79M	0.64M
KXNFL1SEED	Conference 1 seed	32	0.64M	0.56M
KXNFLFFLEADER	Fantasy positional leader	361	0.47M	0.41M
KXLEADERNFLPYDS	Passing yards leader	28	0.28M	0.26M
KXNFLRECYDS	Receiving yards prop	437	0.19M	0.18M
KXNFLRSHYDS	Rushing yards prop	289	0.13M	0.13M

Table 3: Kalshi NFL contract series, September 2026 snapshot of open markets.

We pulled, through the Kalshi REST API, every market in the winner, spread and total series listed since May 2026 (5,402 markets on 354 events), their complete trade tape (12.9 million trades, 1.98 billion contracts), and hourly candlesticks with best bid and ask, volume and open interest (1.17 million candles). The 2025 season is archived out of the public API. Three facts from this data shape the design: 82% of winner-contract volume trades after kickoff; the book sits at the 1-cent minimum tick for nearly all of game day, so the mid-price is a clean quote; and hourly price changes have a standard deviation of about 17 cents in play against 1 cent before the game, so in-play risk is inventory risk rather than liquidity risk. Spread strikes are listed every half point from ± 1.5 to about ± 30 for each team, and totals every half point over roughly ± 20 points around

the line, so a game's ladder pins the market-implied distribution of margin and total at some 45 points. Pre-kickoff prices are well calibrated (Figure 1) and within 1.1 percentage points of the de-vigged sportsbook closing line on average.

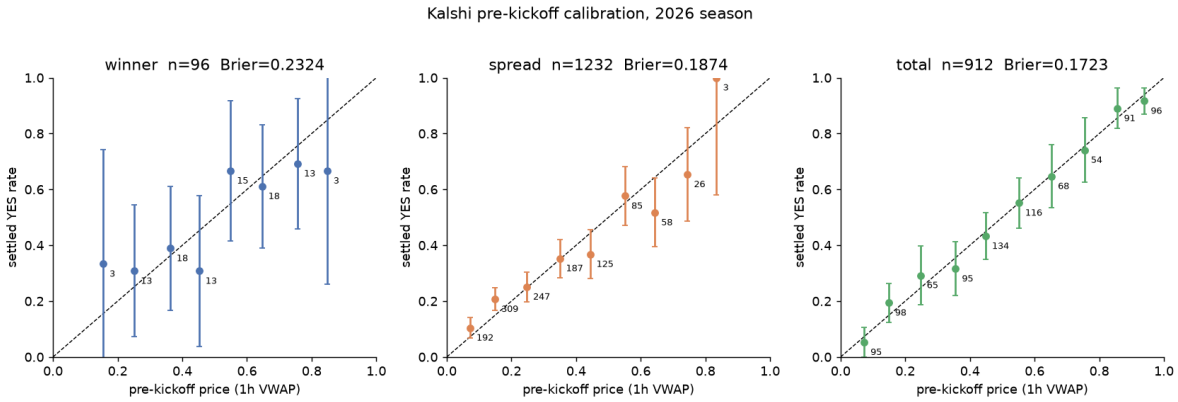


Figure 1: Calibration of Kalshi pre-kickoff prices (1-hour volume-weighted average) against settlement, 2026 weeks 1–3, by contract type.

2.2 NFL results and the pre-game feature table

Historical results, play-by-play, rosters, injury reports and closing lines come from the public nflverse project [15]. From them we built a table of 7,548 games (1999–2026) with 805 columns, one row per game in home-team orientation (Table 4). All data is free.

Group	Cols	Content	From
Team form	604	20 offensive and 20 defensive efficiency statistics from play-by-play (EPA per play, success rate, pass and rush splits, explosive-play rate, yards per play, completion percentage over expected, turnovers, sacks, third-down rate, points per drive, neutral-script EPA), each over five windows (last 4, last 8, exponentially weighted, season to date, previous season), for home, away and the home-minus-away difference	1999; CPOE 2006
Ratings	68	Pre-game Elo (FiveThirtyEight recipe), exponentially weighted margin, opponent-adjusted margin	1999
Quarterback	23	Scheduled starter's weighted EPA per dropback, CPOE, sack and interception rates, career dropbacks, starter-change flag	1999
Injuries	44	Players Out/Doubtful and Questionable by position group; snap share of players now out	2009; snaps 2013
Situational	31	Rest, bye, short week, primetime, dome, surface, temperature, wind, divisional, playoff	1999
Market	10	Closing spread, total, moneylines; stored for benchmarking, never used as features	1999
Targets	5	Home score, away score, margin, total, home win	1999

Table 4: The pre-game feature table.

Every feature is computed from information dated before kickoff: rolling statistics are shifted by one game, Elo is the pre-update value, and injury counts come from the report filed before the game. Following Glickman and Stern [9], each team statistic enters as the home value, the away value and their difference. We test for leakage by perturbation: one game's play-level statistics are multiplied by ten and thirty points added to its score, the table is rebuilt, and the game's own 233 team features and Elo must be unchanged while the two teams' later games change. Six random games pass. The test earned its keep: the nflverse schedule carries two post-game fields, `result` and `total`, that equal the targets. A forest trained with them reported a margin error of 2.4 points with 85% of its importance on those two columns; they were removed and the table re-scanned for any column correlated above 0.6 with a target.

Amine comment: Could you please state when forecasts are made and confirm that weather, starting-quarterback and injury information was available by then. Changing game statistics does not test when these other inputs became known.

Sai comment: Fixed. The paragraph below fixes the forecast time at kickoff and goes through each input: the starting quarterback is the recorded starter (known 90 minutes before kickoff), injuries come from the timestamped final report, and weather is the observed conditions; refitting without the weather columns changes the margin error by 0.001 points (Section 3.2.2).

When the forecast is made. The perturbation test checks that no feature uses the game's own plays or score; it does not check when each input became public. We fix the forecast time at kickoff, the same instant at which the closing line is taken, and list the inputs whose availability depends on that choice. *Starting quarterback*: nflverse records the quarterback who started, which is the information a bettor has when the inactive list is published 90 minutes before kickoff; a forecast made earlier in the week would have to treat a doubtful starter as a probability. *Injuries*: the counts come from the final official injury report, which nflverse timestamps (`date_modified`); it is filed on the Friday or Saturday before a Sunday game, so it is available by kickoff and usually two days earlier. *Weather*: the temperature and wind columns are the conditions recorded at the game, not a forecast, so they are strictly known only afterwards. A game-time forecast would be close but not identical. To bound what this costs, we refit the forest walk-forward with the four weather columns removed; the margin error changes by 0.001 points and the total error by 0.013 (Section 3.2.2). *Rolling form, Elo, rest, venue* are known days in advance. No feature uses the betting line.

2.3 Testing

All programs are implemented in Python in a shared repository and run locally or on Google Colab. Models are evaluated walk-forward by season, training on all seasons before t and testing on season t, because games within a season are related through the teams.

3 Weekly progress

3.1 Report of 30 September 2026: pricing as outcome modelling

Summary. Framed pricing as outcome modelling (Section 1.1); surveyed the sports-pricing literature; pulled Kalshi markets, trades and candlesticks and nflverse results (Section 2); explored the empirical score distribution; built a worked example of an empirical distribution positioned to a game and priced against a live order book.

3.1.1 What the empirical score distribution looks like

Three findings from 3,076 games (2015–2026) constrain the shape of $p(h, a)$ (Figure 2). The absolute margin lands exactly on 3 in 14.8% of games and on 7 in 8.7%, 2.9 and 1.9 times what a normal distribution centred on the point spread implies; a smooth model misprices spread strikes near these "key numbers" by 5 to 10 cents. Margin and total are nearly uncorrelated (0.03) but share parity in every game, since they are the difference and sum of the same two integers, so half of the margin-by-total lattice is empty and a copula on the two marginals would put mass there. And the closing spread is an unbiased forecast with a residual standard deviation of 12.7 points, slightly tighter than the 13.86 of Stern [19].

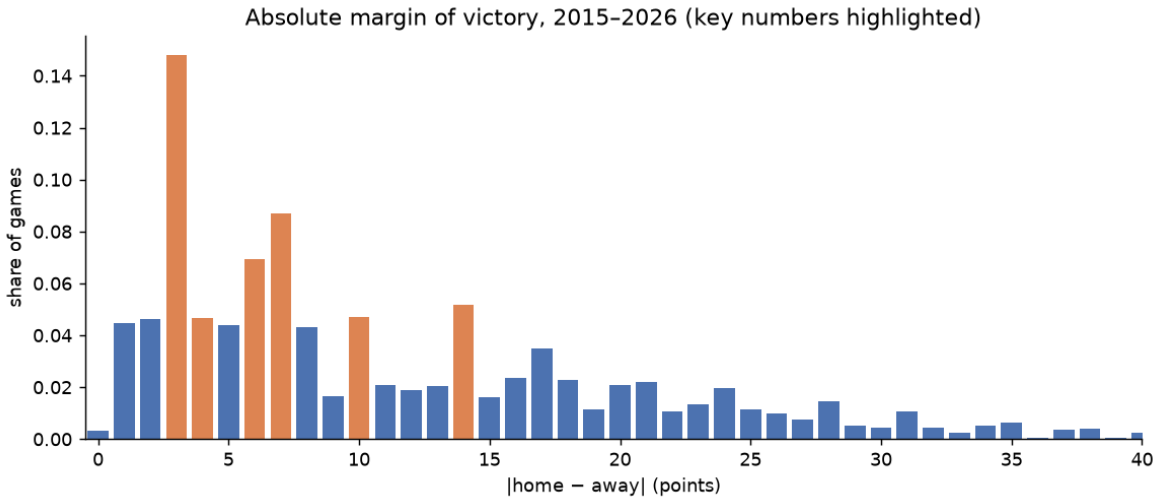


Figure 2: Absolute margin of victory, 2015–2026 (3,076 games); key numbers 3, 7, 6, 10 and 14 in orange.

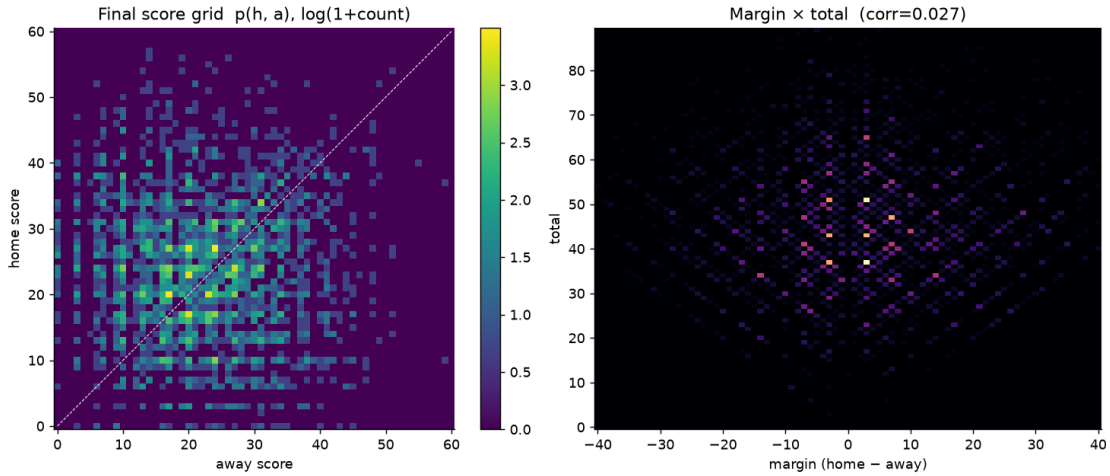


Figure 3: Left: the empirical final-score grid $p(h, a)$ (log counts). Right: margin against total; the chequerboard is parity.

3.1.2 Positioning the empirical distribution for a game

We want a distribution p that (i) has the game's expected margin and total and (ii) is otherwise as close as possible to the historical prior p_0. Minimising the Kullback–Leibler divergence $\mathrm{KL}(p \,\|\, p_0)$ subject to the two moment constraints gives the exponential-family solution

$$p(h,a) \;\propto\; p_0(h,a)\,\exp\!\big(\lambda_1(h-a)+\lambda_2(h+a)\big), \tag{1}$$

with (λ_1,λ_2) solved so that $\mathbb{E}[h-a]$ and $\mathbb{E}[h+a]$ match the targets, a two-equation root find. Because the tilt only reweights cells, it cannot create mass where the prior has none: a 3-point win stays 2.9 times as likely as a 4-point win. A Poisson or normal model would have to be taught key numbers, parity and cell clustering; the empirical grid already has them.

Amine comment: Could a possible score pair receive zero probability simply because it is absent from the historical sample? How such outcomes are handled and how smoothing affects extreme-outcome probabilities and expected shortfall.

Sai comment: Fixed. The paragraph and Table 5 below quantify it: 19–31% of a season's games land on an unseen exact pair, the 2% pseudo-count fills every cell whose two scores have been seen, and moving it from 0 to 10% shifts tail probabilities by hundredths of a point and a deep strike's ES_{99} by under 0.5%.

Score pairs never seen. The reweighting property cuts both ways: a reachable score pair that is absent from the sample would keep probability zero under any tilt. This is not rare. Of the roughly 6,400 reachable cells (either score 0–80 except 1), the 3,028 games of 2015–2025 occupy 1,065, and between 19% and 31% of a season's games end on an exact pair that had not occurred in the preceding seasons since 2015 (31% in 2020, 19% in 2023, 22% in 2025). We therefore add a pseudo-count: a mass of 2% spread over the cells in proportion to the product of the two marginal score distributions, which fills every cell whose two scores have each been seen (2,850 cells) and leaves at zero only combinations involving a score never recorded. The exact pair matters for a log-likelihood but not for a contract, since every contract is a sum over a region of the grid and the regions near the tails contain many cells. Table 5 shows the effect of the pseudo-count on the tilted grid of a typical line: tail probabilities move by a few hundredths of a percentage point and the 99% expected shortfall of a deep strike by less than 0.5%. The choice of 2% is a convention; moving it to 10% changes the same quantities by about one part in fifty, with the largest effect on the price of the deepest strikes, which is where the sample is thinnest and a model-based tail would be the better tool.

Pseudo-count mass	**0**	**2% (used)**	**10%**
Non-zero cells	1,065	2,850	2,850
$P(\lvert h-a\rvert \geq 21)$	16.53%	16.51%	16.42%
$P(\lvert h-a\rvert \geq 28)$	6.86%	6.83%	6.72%
$P(\lvert h-a\rvert \geq 42)$	0.345%	0.344%	0.340%
$P(h+a \geq 70)$	4.07%	4.06%	4.05%
Price of "home wins by > 27.5"	1.95¢	1.94¢	1.90¢
ES_{99}, short 1,000 of that contract	\$980.5	\$980.6	\$981.0
ES_{99}, short 1,000 of "over 69.5"	\$959.3	\$959.4	\$959.5

Table 5: Effect of the pseudo-count on the 2015–2025 grid tilted to home −3, total 44.5.

3.1.3 Worked example: Pittsburgh at Cleveland

Pittsburgh at Cleveland, 1 October 2026, with the live Kalshi order book captured the day before (46 contracts). The sportsbook line was Cleveland -2.5 and total 38.5. Tilting the 3,076-game prior to those two numbers gives $\lambda_1 = -0.0222$, $\lambda_2 = -0.0399$ (Figures 4 and 5). Table 6 compares model prices with the Kalshi mid for every contract.

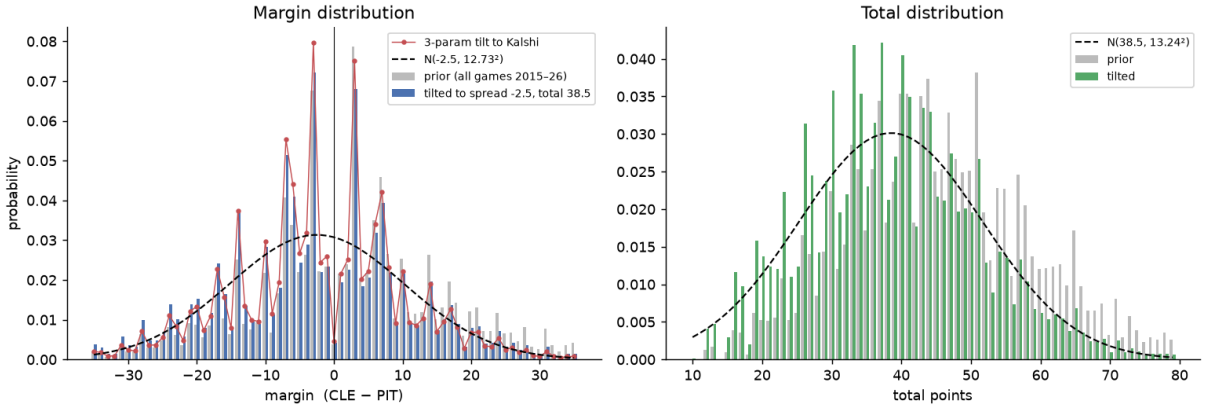


Figure 4: Margin (left) and total (right) distributions: grey the prior, blue the prior tilted to the book line, red a three-parameter tilt calibrated to Kalshi, dashed the normal.

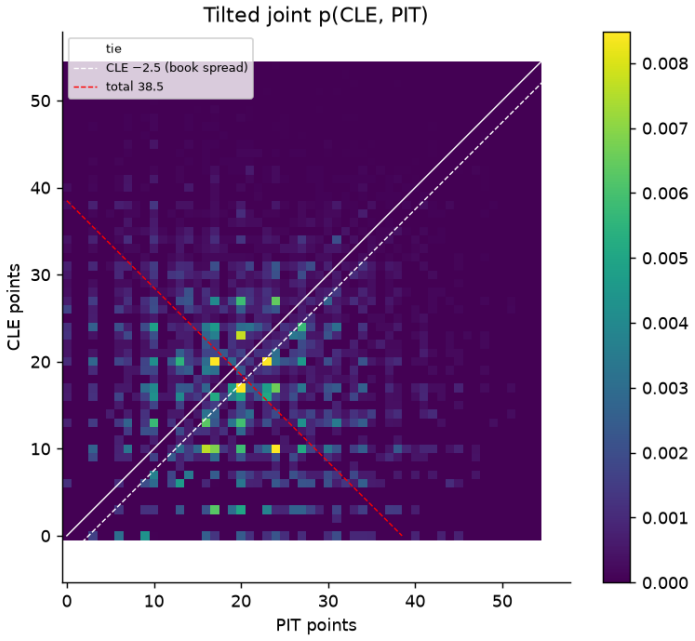


Figure 5: The tilted joint table for Pittsburgh at Cleveland. The diagonal is a tie, the dashed line the book spread and the red line the total; each contract is a sum over one side of one of these lines.

Model	All 46	Winner	Spread strikes	Total strikes
Untilted prior	11.2	12.5	8.7	14.3
Normal around the line	1.89	0.28	2.32	1.48
Tilt to the book line (no Kalshi data)	1.52	0.08	1.98	1.06
Two-parameter tilt fitted to Kalshi mids	1.49	0.76	2.00	0.88
Three-parameter tilt (adds margin^2) fitted to Kalshi	0.76	0.24	0.64	0.99

Table 6: Mean absolute error in cents of model price against the Kalshi mid, Pittsburgh at Cleveland. Quoted bid–ask spreads were 1–2 cents.

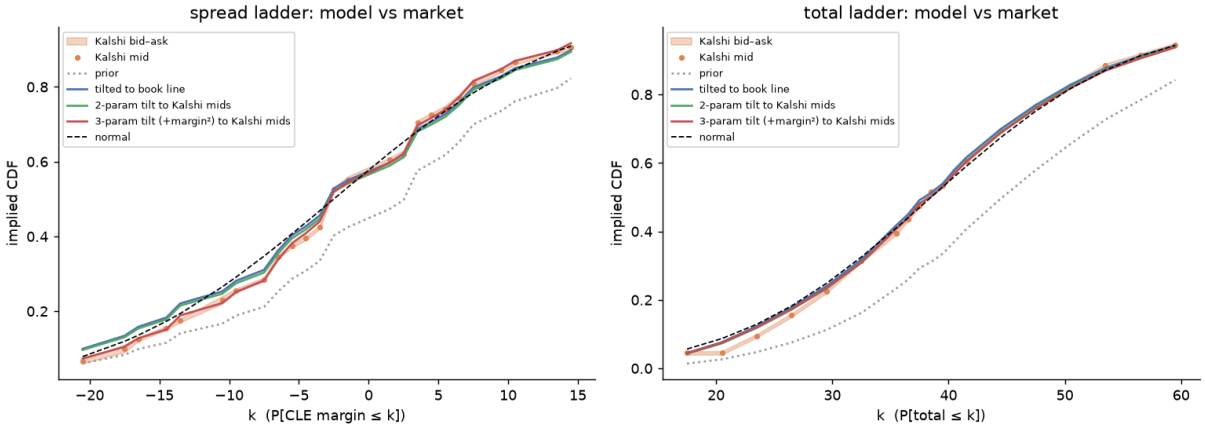


Figure 6: Market-implied cumulative distributions from the spread (left) and total (right) ladders against the model variants. The shaded band is the Kalshi bid–ask.

Three lessons follow. Locating the distribution with the book line removes 87% of the pricing error and prices the winner contract to 0.1 cents. Discreteness matters on the ladder: the normal is off by up to 6 cents on a single strike. And two moments do not control the tails: the book-tilted grid overprices Pittsburgh blowouts by 2–4.5 cents because the prior's margin has standard deviation 14.2 while the market prices this low-scoring game nearer 12.3. A third parameter on $(h-a)^2$ lets the market set the dispersion and the tail error disappears. This is the strongest argument for a model that predicts dispersion as well as location, and it motivates the score model of Section 3.2.2.

3.1.4 What moves a game's expected score

Before building features we listed the factors practitioners weigh when setting a line (Table 7); the feature table of Section 2 covers all but travel distance.

Factor	Typical effect	Notes
Starting quarterback	±3 to 10 pts	The largest single factor; elite starter versus backup is worth 7–10 points or 20–25 percentage points of win probability. Status should be modelled as a probability when questionable.
Team strength	baseline	Elo or power ratings, or EPA per play and success rate on offence and defence; adjust for opponent and regress toward the mean early in the season.
Home field	1.5 to 2 pts	Has shrunk over the past decade; varies by venue and travel.
Other key injuries	0.5 to 2 pts each	Costliest at offensive tackle, edge rusher, top cornerback and top receiver; multiple injuries on one unit compound.
Rest and schedule	0.5 to 1.5 pts	Bye week helps; short week and long road trips hurt; time-zone changes matter for early-kickoff West Coast teams.
Weather and venue	totals	Wind above 15 mph and heavy precipitation suppress scoring; lowers the total and compresses variance. Domes remove it.
Coaching	0.5 to 1 pt	Fourth-down aggressiveness, clock management, in-game adjustments.
Motivation and context	small, noisy	Playoff seeding, eliminated teams and week-18 rest can matter in specific spots.
Turnover luck	shrinkage	Fumble recoveries and close-game records are largely noise; use them to shrink apparent strength.

Table 7: Factors in a game's expected score, with typical magnitudes as cited by practitioners.

Conclusion of the week. It is better to build the scoring model first: predict the two absolute scores, home and away, from pre-game features, and let the score grid turn them into prices. Professors Amine and Shasha suggested dimensionality reduction (PCA, UMAP), a multi-output random forest, LARS and cascade-correlation networks, a multi-output linear baseline, and a neural network parameterising a bivariate Gaussian $(\mu_h, \mu_a, \sigma_h, \sigma_a, \rho)$; Professor Shasha noted that random forests, especially with boosting, handle many features without reduction and discard what is useless.

3.2 Report of 7 October 2026: literature, feature table, two-output random forest

Summary. The question after the first report was whether anyone has predicted NFL scores before, with what data, and how. This week: a literature survey; a catalogue of the data used in the literature against what nflverse provides; the 805-column pre-game feature table with its leakage test (Section 2); and a first two-output random forest evaluated walk-forward over sixteen seasons.

3.2.1 Literature

The first question raised after the 30 September report was whether anyone has tried to predict NFL scores before, what data they used, and how they thought about the problem. The literature falls into five groups.

Rating models from scores alone. Harville [12] fitted a mixed linear model to score differences from 1,320 games and obtained an average error of 10.68 points against 10.49 for the bookmaker. Glickman and Stern [8] let each team's strength drift from week to week and season to season in a Bayesian state-space model and matched the Las Vegas line on a 1993 test set. Their handbook chapter [9] surveys these methods and recommends entering each covariate as the difference and the average of the two teams' values, which we adopted. FiveThirtyEight's Elo system [6] is the practical version: one strength number per team updated from margin of victory, with adjustments for home field, rest and the starting quarterback. All of these use only results and the schedule, and all land a few tenths of a point behind the line.

Exact scores and distributions. Baker and McHale [2] is the only paper that forecasts both teams' exact NFL scores, with a birth-process model and pre-game covariates; it too performs about as well as the bookmaker. Stern [19] showed that margin minus spread is approximately normal with standard deviation near 14, which gives the standard conversion from a spread to a win probability; Stern [20] and Polson and Stern [16] extend this to the progress of a game and to a market-implied volatility. In soccer, Dixon and Coles [5] model scores as correlated Poisson counts and Divos et al. [4] price in-play bets by risk-neutral methods; the lesson for us is that football scoring is discrete in a way Poisson models do not capture.

Play-level information. Yurko, Ventura and Horowitz [22] defined expected points and win probability from play-by-play data and showed how to attribute them to plays and players. Expected points added per play is the efficiency measure the analytics community has settled on; the nflverse project provides it for every play since 1999 and it is the source of most of our team-form features.

Market efficiency. Levitt [13] showed from a bettor-level dataset that bookmakers do not balance their books but set prices as skilled forecasts that exploit bettor biases, which is why the closing line is hard to beat. Boulier and Stekler [3] and Song, Boulier and Stekler [18] compared statistical systems, experts and the betting market and found the market most accurate, with models and experts tied. Zuber, Gandar and Bowers [23] and Gray and Gray [10] found only small, unstable inefficiencies. Lopez, Matthews and Baumer [14] go the other way and use moneylines as the best available measure of team strength. Shin [17] and Wolfers and Zitzewitz [21] show how to read probabilities from odds and from prediction-market prices.

Machine-learning studies. A systematic review by Galekwa et al. [7] and a number of student projects report accuracies of 70 to 95 percent at picking winners. Where the methods are described, these come from same-game statistics (leakage) or from very small test sets. The closing line picks the winner about 66 percent of the time; nothing credible is far above that. We took two lessons: build an explicit leakage test, and do not expect a flexible learner to add much over a well-specified rating model with 7,000 games.

Data in the literature. The serious score models use only scores, dates and home field; richer inputs appear in three families: play-level efficiency, quarterback identity and betting lines. Bet-volume data (Levitt; Paul and Weinbach) is proprietary; the Kalshi trade tape is the free equivalent. No paper reviewed uses injuries, snap counts or depth charts in a score model, although Glickman and Stern [9] name player-level factors as the obvious gap. Everything the literature uses is available free from nflverse except opening lines and bet splits.

Implication for our design. Point accuracy for NFL scores is capped near the closing line for every published method. The sensible goal is a calibrated distribution over both scores, consistent across every derived quantity, rather than a lower point error. The suggestions received from Professors Amine and Shasha fit this picture: a multi-output random forest can use every feature without selection, a linear baseline shows whether the forest adds anything, and a probabilistic extension predicting two means, two standard deviations and a correlation is the natural destination for a two-score model.

3.2.2 Predicting both teams' scores

Why two scores. The object the engine needs is the joint distribution over (h, a). Predicting each team's score is the natural first step: the two predictions locate the centre of that distribution and the spread of the residuals around them gives its width. A model for the margin alone discards the level of scoring; a 30–27 game and a 17–14 game have the same margin and very different distributions of totals and of every strike.

Models and evaluation. Table 8 lists the models. All use the 759 non-market features of Section 2; the closing line is a benchmark only. Evaluation is walk-forward: train on every season before t, test on season t, for $t = 2010, \ldots, 2025$ (4,363 test games).

Model	Inputs	Role
League average	Mean scores in the training seasons	Floor
Elo	Results only; implied spread is Elo difference / 25	Rating baseline
Two-output ridge regression	759 features, median-imputed, standardised, penalty by cross-validation	Linear baseline
Two-output random forest	759 features, 500 trees, at least 20 games per leaf, 30% of features per split, no tuning	Main model: predicts (h, a) jointly
Closing line	Consensus sportsbook spread and total	Benchmark

Table 8: Models compared. The forest predicts the two scores as a pair: each leaf holds the mean home and away score of similar past games.

Model	Home score	Away score	Margin	Total
League average	8.09	7.91	11.18	11.02
Elo	–	–	10.38	–
Two-output ridge	7.62	7.40	10.34	10.63
Two-output random forest	7.63	7.45	10.35	10.66
Closing line	–	–	**10.05**	**10.47**

Table 9: Mean absolute error in points, walk-forward 2010–2025.

Point accuracy. The forest explains 10.8% of the variance of the home score and 9.3% of the away score out of sample. These numbers are normal for the problem: a single game is mostly noise around a modest signal, and the closing line, with every input a bettor can find, explains only a

little more. The forest and the ridge are indistinguishable at every target. On the margin they are 0.03–0.04 points better than Elo and 0.3 points worse than the line (Figure 7); the line has the lowest error in 15 of 16 seasons. Straight-up winner accuracy is 64.7% for the forest, 64.6% for ridge, 64.5% for Elo and 66.3% for the line. Against the spread the forest picks the covering side 49.9% of the time, as an efficient line implies.

Amine comment: Please report uncertainty for the differences in error on the same games, allowing for dependence over time, and clarify forecast timing relative to the closing line. The table shows gains over Elo of 0.03–0.04 points, not a quarter point.

Sai comment: Fixed. The "quarter of a point" is corrected to 0.03–0.04 above; the paragraph and Table 10 below give paired error differences with i.i.d., week-clustered, season-clustered and Newey–West standard errors (gain over Elo about one standard error; gap to the line seven to eight), and state that model and line are both taken at kickoff.

Because every forecast is scored on the same 4,363 games, the differences between them are paired, and their uncertainty has to allow for the fact that games in the same week and season are related. Table 10 gives the mean difference in absolute margin error for each pair with four standard errors: treating games as independent, clustering by week (271 clusters), clustering by season (16 clusters), and a Newey–West estimate over the game sequence with a lag of four weeks. The four agree closely, which says the game-level errors are nearly uncorrelated once the forecasts are differenced. The gain over Elo, 0.03–0.04 points, is about one standard error and shows up in only 9 or 10 of the 16 seasons: it is not distinguishable from zero. The gap to the closing line, 0.29–0.33 points, is seven to eight standard errors and holds in 15 of 16 seasons. The forest and the ridge differ by 0.01 ± 0.03. On timing, every model forecast uses only information available at kickoff and none uses the line (Section 2); the closing line is also the kickoff price, so the two are compared at the same instant. The weather columns are the one input strictly observed rather than forecast. Refitting the forest walk-forward without the four weather columns (`scripts/model_rf_noweather.py`) changes the margin error from 10.352 to 10.353 and the total error from 10.663 to 10.676, so nothing in the comparison rests on knowing the weather in advance.

Difference in MAE	Mean	SE i.i.d.	SE week	SE season	SE NW	t, 16 seasons	Seasons better
Forest − Elo	−0.030	0.029	0.028	0.032	0.029	−0.9	10 of 16
Ridge − Elo	−0.041	0.039	0.040	0.034	0.036	−1.2	9 of 16
Forest − Ridge	+0.011	0.029	0.031	0.022	0.028	0.5	6 of 16
Forest − Line	+0.298	0.043	0.041	0.039	0.040	7.5	1 of 16
Ridge − Line	+0.287	0.043	0.042	0.038	0.040	7.4	1 of 16
Elo − Line	+0.328	0.045	0.045	0.040	0.042	8.0	1 of 16

Table 10: Paired differences in absolute margin error, 4,363 games, 2010–2025. Negative means the first forecast is better. The t statistic is the season-level paired t on the 16 seasonal means; the last column counts seasons in which the first forecast had the lower error.

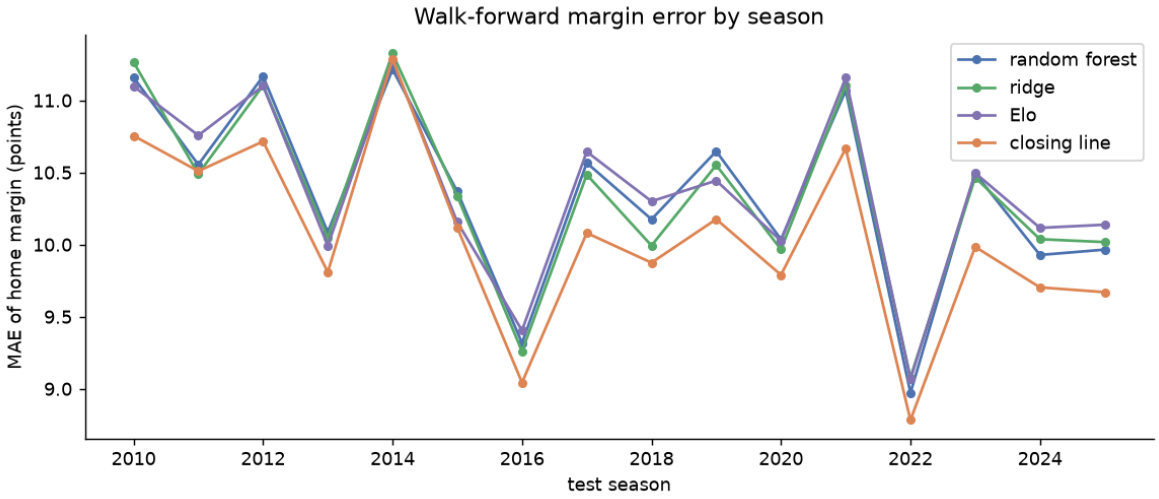


Figure 7: Margin error by test season. All methods share the hard and easy seasons; the closing line is lowest in 15 of 16.

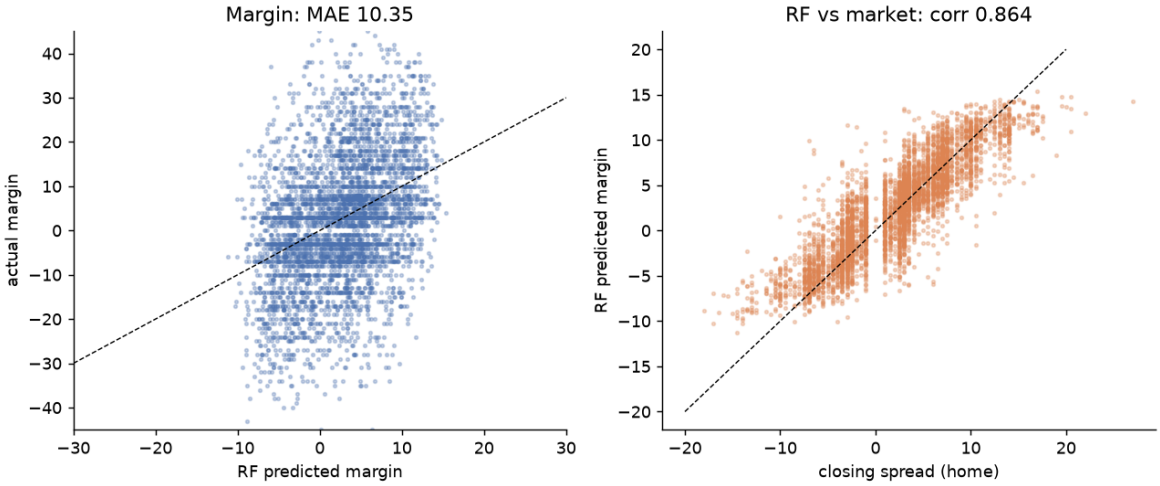


Figure 8: Left: predicted against actual margin. Right: the forest's margin against the closing spread (correlation 0.86).

The two-output structure. Because the forest predicts a point in the (h, a) plane, its residuals describe a two-dimensional distribution around that point (Table 11, Figure 9). The correlation between the home and away residuals is 0.01: once the two means are set, the leftover error in one score says nothing about the other. Residual spread grows with the predicted score, so high-scoring games are less predictable in points, as a Poisson-like scoring process implies. Calibration per score is good: when the forest predicts 20 points the team scores 20 on average, across the range 15 to 33.

Quantity	Home	Away
Standard deviation of the actual score	10.21	9.83
Standard deviation of the residual	9.64	9.36
Range of predicted scores	15.8 to 33.3	13.3 to 30.4
Residual s.d., lowest third of predictions	9.20	8.88
Residual s.d., highest third of predictions	10.11	9.76

Table 11: Residual structure of the two-output forest, 4,363 out-of-sample games. Residual correlation 0.011.

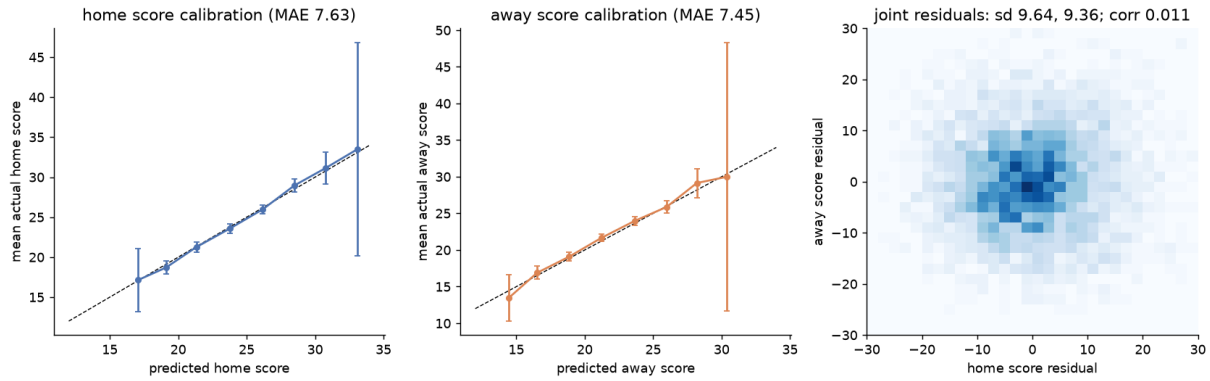


Figure 9: Calibration of the home and away score predictions and the joint distribution of the two residuals.

These residual statistics are the parameters $(\sigma_h, \sigma_a, \rho)$ of the probabilistic extension suggested by Professor Amine. Table 11 and Figure 9 pool the residuals of all sixteen test seasons, which is the right thing for describing them but not for scoring probabilities on those same seasons. For scoring we refit them walk-forward: the probability for a game in season t uses the residual standard deviations and correlation of the out-of-sample predictions for seasons 2010 to $t-1$ only, so the first scored season, 2016, uses six seasons of residuals (1,602 games) and 2025 uses fifteen (4,078). The margin standard deviation estimated this way runs from 13.87 (for 2016) down to 13.40 (for 2025), against 13.36 pooled. Used as a bivariate normal around the two predicted scores, the fold-wise residuals give a home-win probability with Brier score 0.2208 over 2016–2025 (0.2207 with the pooled values), against 0.2220 for Elo and 0.2108 for the de-vigged moneyline. The empirical grid used for the alternative, placing the two predicted means on the score table of Section 1.1, was already fold-wise: its prior for season t is the table of all games before t, and it gives 0.2213. The probabilities are calibrated (Figure 10); the gap to the market is in sharpness, not bias.

Amine comment: Please confirm that residual covariance and the empirical grid use only earlier games in each test fold. Their stated estimation periods overlap the evaluation years.

Sai comment: Fixed. The grid prior was already fold-wise (all games before the test season). The residual covariance was pooled; it is now refit walk-forward from seasons before each test season, as the paragraph above reports (Brier 0.2208 fold-wise against 0.2207 pooled), and the pooled table and figure are labelled as descriptive.

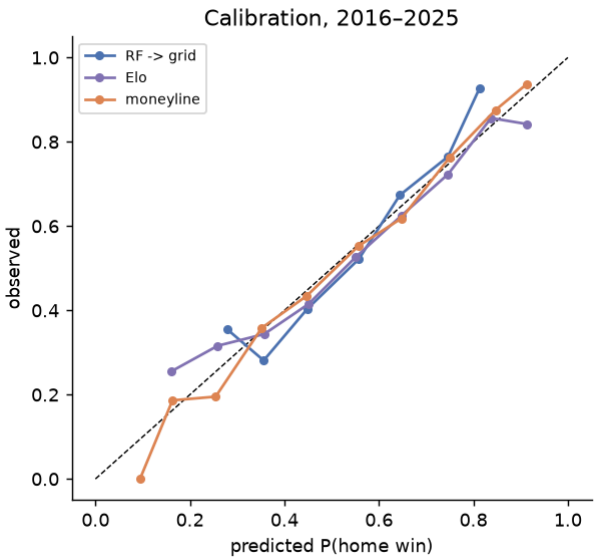


Figure 10: Calibration of home-win probabilities, 2016–2025.

What the forest uses. Averaged over the sixteen fits, the five most important features are all team-strength summaries from results: the difference in opponent-adjusted weighted margin, the difference in weighted margin, and the three Elo features (Figure 11). The first play-by-play feature is yards per play, followed by the starting quarterback's success rate and EPA per dropback and points per drive. Injury, rest and weather features are far down the list. Most of what the model knows is how good each team has been; the efficiency and quarterback features refine that at the margin. The forest used all 759 features with no selection or tuning and matched a regularised linear model, which is what Professor Shasha expected.

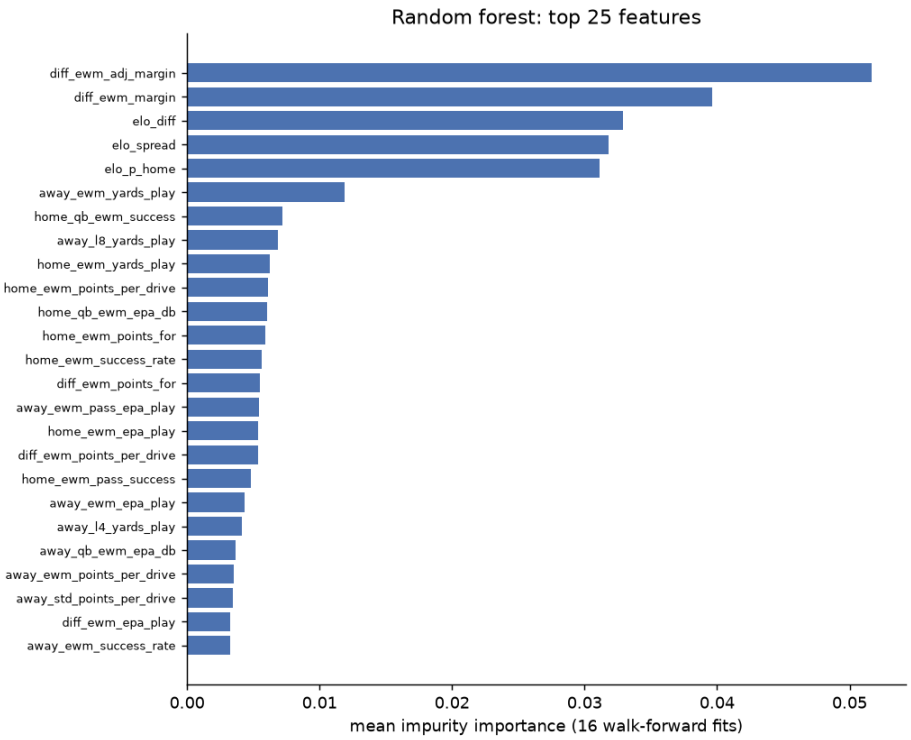


Figure 11: Top 25 features by mean impurity importance over the sixteen walk-forward fits.

3.3 Report of 21 October 2026: ensembles and full distributions

Summary. Two questions from Professor Shasha: can an ensemble of the models we have, or can obtain, do better; and can the models give a distribution for the margin and the total, compared with the Vegas lines.

3.3.1 Can an ensemble do better?

Professor Shasha asked whether an ensemble of the models we have, or can obtain, improves on them. We stacked by least squares, fitting weights on earlier seasons' out-of-sample predictions and applying them to the next season, 2013–2025. To our three models we added two public forecasts, nfelo [11] (a public Elo variant with a quarterback adjustment; we use its rating difference before any regression toward the market) and FiveThirtyEight's Elo (available to 2020), and the sportsbook *opening* line, which the nfelo files carry (Table 12, Figure 12).

Forecast	Margin MAE	Note
Random forest (ours)	10.22	
Ridge (ours)	10.20	
Elo (ours)	10.25	
Stack of our three	10.18	weights 0.19 / 0.36 / 0.51
nfelo, pure model	10.01	history back-computed with the current model, so optimistic
Stack: ours + nfelo	10.04	our models receive weight near 0
Opening line	10.02	
Closing line	9.92	
Stack: ours + nfelo + closing line	9.95	closing line 0.58, nfelo 0.72, ours near 0

Table 12: Stacking test, home margin, 2013–2025 (3,562 games with all forecasts available).

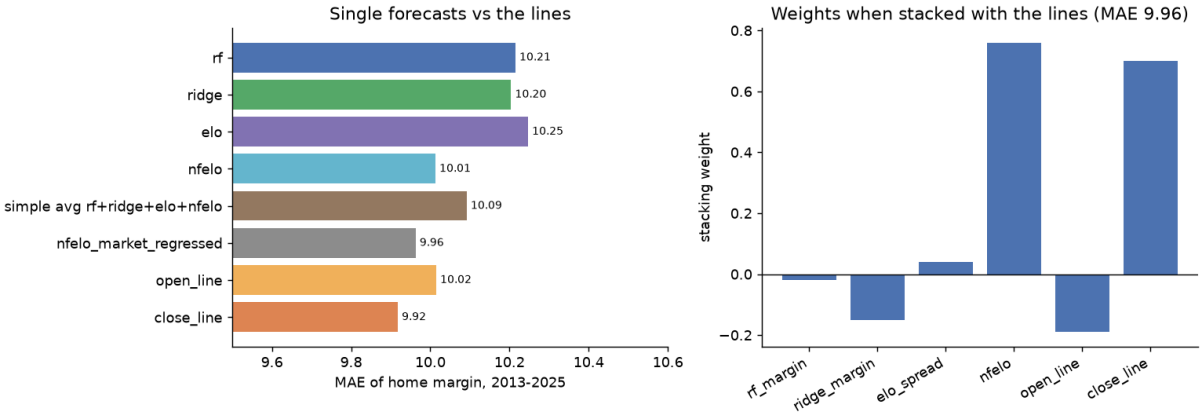


Figure 12: Left: single forecasts and the lines. Right: stacking weights when our models, nfelo and both lines are combined.

Averaging our own models gains only a few hundredths of a point, because their predictions are correlated at 0.89–0.94. nfelo is a better single forecast than anything we built, by about 0.2 points, which we attribute to its quarterback adjustment and to its published history being generated by the current version of the model. Most important, no stack improves on the closing line: whenever the line is a component it takes nearly all the weight. The same holds for totals, where the opening total (10.47) is in fact slightly better than the closing total (10.53) in this sample. Ensembling what we have does not beat the market; an ensemble that could would need forecasts carrying information the market does not yet have.

3.3.2 Full distributions and the Vegas lines

The second question was whether the models can give a distribution for the margin and the total and how it compares with the lines. The two predicted scores set the centre, and the distribution around them comes either from the bivariate normal with the residual covariance or from the empirical grid tilted so that its mean margin and total match the forest (Figure 13). From either, the probability of any outcome is a sum: P(home wins), P(home covers the spread), P(over the total).

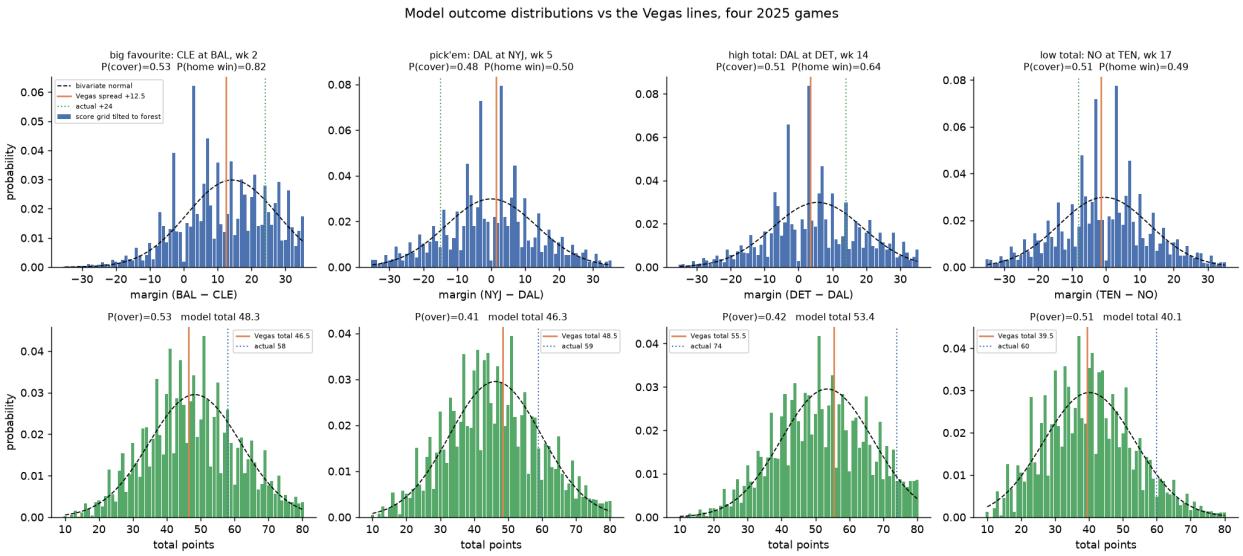


Figure 13: Four 2025 games. Top: margin distribution from the tilted score grid (bars) and the bivariate normal (dashed), with the Vegas spread (orange) and the actual margin (dotted). Bottom: the same for the total.

How the calibration is measured. For every game of 2016–2025 the empirical grid, with its prior built from the seasons before the test season, is tilted so that its mean margin and total equal the forest's two predictions; the model's probability of an event is the mass of that grid on the event's region. Games are then grouped by predicted probability in bins of width 0.1, and the observed frequency of the event in each bin is plotted against the mean prediction (Figures 14 and 15). A forecast is calibrated if the points lie on the diagonal; it is informative if the predictions spread out along it. The Brier score summarises both, and we compare it with the climatological forecast that assigns every game the base rate.

Thresholds set by the market. Across all 2,758 games the model's $P(\text{cover})$ averages 0.50 with a standard deviation of 0.09 and $P(\text{over})$ averages 0.49 with a standard deviation of 0.08; only a quarter of games get a cover probability outside 0.40–0.60. The calibration of these probabilities against outcomes is flat (Figure 14): when the model says 65% to cover, teams cover about 52%, and the regression slope of outcome on prediction is 0.06 for cover and 0.11 for over. Brier scores of 0.257 for cover and 0.255 for over are slightly worse than the 0.250 of the base rate. So, relative to the threshold the market has chosen, the model's probabilities carry no information. This is not a failure of calibration in the usual sense; it is what an efficient line implies. The closing spread is the market's median for the margin, set with everything the market knows, so conditioning on it removes exactly the part of the model's information that the market already has, and what remains is noise.

Amine comment: Please clarify how calibration is done. When the model predicts a 65% chance of covering, the observed rate is about 52%. Please also test fixed thresholds chosen in advance, separately from thresholds set by the market.

Sai comment: Fixed. The paragraph "How the calibration is measured" above states the procedure. The paragraph "Thresholds fixed in advance" below, with Table 13 and Figure 15, tests seven pre-chosen events: the model has 8–11% Brier skill on margin events and 3–4% on totals with

slopes 1.13–1.27, against slopes of 0.06–0.11 at the market's thresholds.

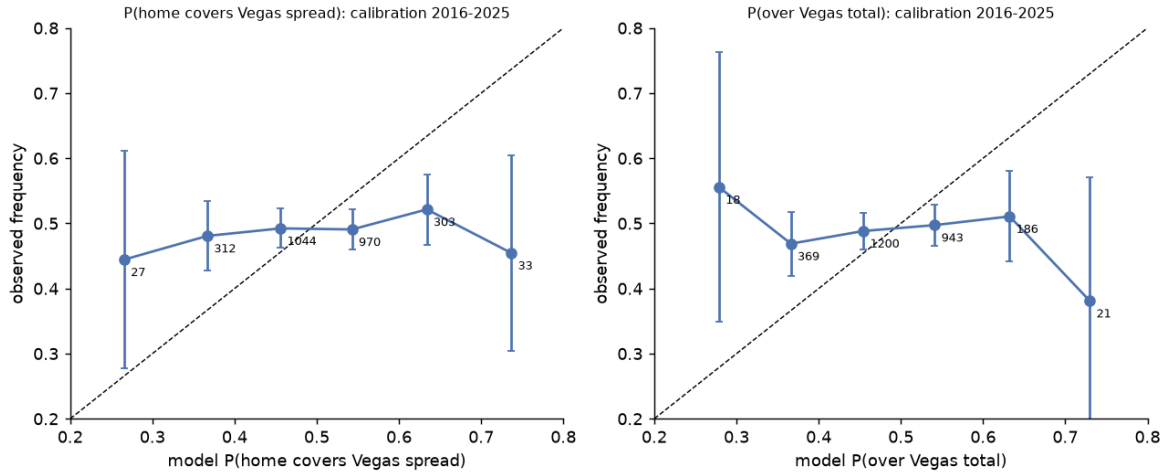


Figure 14: Calibration of the model's cover and over probabilities against the closing line, 2016–2025; bin sizes shown.

Thresholds fixed in advance. The test Professor Amine asked for separates the two questions. We fixed seven thresholds before looking at the data, chosen to sit at round numbers and key numbers rather than at any game's line: the home team wins; wins by more than 3; by more than 7; the away team wins by more than 3; and the total exceeds 41, 45 and 48. For each the model probability is read from the same tilted grid, and the outcome is scored against the same 2,761 games (Table 13, Figure 15). At every fixed threshold the probabilities are informative: the Brier score beats the base rate by 8–11% on the margin thresholds and 3–4% on the totals, and the predictions spread across a range of 0.2 to 0.8. They are close to calibrated, with a regression slope of outcome on prediction between 1.13 and 1.27: the model is slightly under-confident, pushing its probabilities toward one half a little more than the outcomes warrant, which is the usual behaviour of a forest with large leaves. The totals carry less information than the margins (the predictions have a standard deviation of 0.08 against 0.13), consistent with the point-error table, where the model's total is further behind the line than its margin.

Event	Base rate	SD of p	Brier	Brier, base rate	Skill	Slope
Home wins	0.550	0.132	0.2212	0.2475	10.6%	1.25
Home wins by > 3	0.424	0.131	0.2182	0.2442	10.7%	1.27
Home wins by > 7	0.303	0.122	0.1915	0.2111	9.3%	1.18
Away wins by > 3	0.332	0.122	0.2032	0.2219	8.4%	1.13
Total > 41	0.595	0.076	0.2330	0.2409	3.3%	1.20
Total > 45	0.481	0.079	0.2406	0.2497	3.6%	1.24
Total > 48	0.396	0.077	0.2312	0.2391	3.3%	1.16
Home covers the closing spread	0.493	0.088	0.2568	0.2499	−2.8%	0.06
Over the closing total	0.490	0.080	0.2550	0.2499	−2.0%	0.11

Table 13: Model probabilities from the tilted grid scored against outcomes, 2016–2025 (2,761 games; ties excluded where they occur). Skill is one minus the ratio of the Brier score to that of the base-rate forecast; slope is the least-squares slope of the outcome on the prediction, 1 for a calibrated forecast.

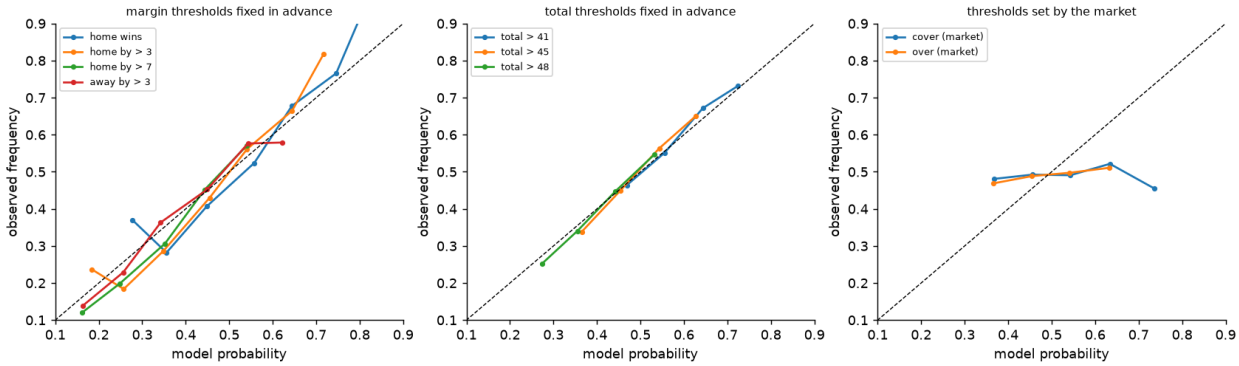


Figure 15: Reliability of the model's probabilities at thresholds fixed in advance (left, centre) and at the thresholds set by the market (right), 2016–2025; bins with at least 30 games.

The two tests together give the clean statement. The distributions are calibrated and informative for any event defined in advance, and carry nothing beyond what the closing line already contains for the event the line defines. The engine's value therefore lies in pricing the whole distribution consistently, not in picking sides against the market.

3.4 Report of 28 October 2026: parlays

Summary. A parlay is a bet that two or more things all happen, for example "Chiefs win and Eagles win". It pays only if every leg wins. Since we can already price single-game contracts, the question this week was whether we can price parlays, and whether the probabilities we build ourselves are good enough to do it. We restricted ourselves to parlays on match winners (moneyline legs); spreads and totals come later. Two things were done: a historical test over eight seasons of what multiplying single-game probabilities gives, and a test on 854,296 Kalshi parlays from the first eight weeks of 2026 of how Kalshi prices them and how they settled.

3.4.1 How we thought about it

If the games in a parlay do not affect each other, the chance that all legs win is the product of the chances of each leg. So a parlay price has two ingredients: a probability for each game, and the assumption that the games are independent. We test both. For the single-game probability we compare four sources:

1. **The closing line.** The sportsbook's moneyline just before kickoff, converted to a probability. A price of -500 means bet \$500 to win \$100, which implies a win chance of $500/600 = 83.3\%$. The two sides of a game add to more than 100% because the book charges a fee; we scale them so they add to 100% ("de-vigging").
2. **Our model, raw.** The two-output random forest predicts both teams' scores; placed on the empirical score grid (Section 1.1) this gives a win probability.
3. **Our model, recalibrated.** The same probability corrected for how it did in past seasons (isotonic regression fitted on earlier seasons' out-of-sample predictions only, so nothing from the future is used).
4. **Elo.** A rating built from wins and losses alone. Each team has a number; a win moves points from loser to winner, more for an upset. The gap between two ratings, plus a home-field allowance, converts to a win probability by $1/(1 + 10^{-\text{gap}/400})$.

3.4.2 Data

History. Every regular-season game from 2018 to 2025 with a closing moneyline and an out-of-sample model prediction (2,219 games). Within each week we formed every pair of games with every choice of side (60,384 two-leg parlays), and random samples of three- and four-leg combinations (338,336 and 340,000). For each parlay we computed its probability under each source by multiplying, and recorded whether it hit.

Kalshi. Kalshi builds parlays on request as "multivariate event" markets: the user picks legs, the exchange creates a market that lists them, and it settles yes only if every leg does. These markets are not in the three game series we had been using. They live in "cross-category" series that close thousands of untraded quotes a minute, so they cannot be read in full; instead we read only the forty-minute windows around each NFL game's settlement, since a parlay closes within minutes of its last game (59 windows over the 2026 season). From about 13 million markets read, we kept those whose every leg is an NFL contract, 1.85 million in all, and for this report those whose every leg is a match winner: 854,296 markets holding 4.7 billion contracts, of which 20,367 (2.4%) settled yes. People buy long parlays: the median has eight legs, only 7,062 have two and 25,026 three. Each market carries its legs, its last traded price and its volume. The fills themselves are not on the public trade tape; they appear in the hourly candlesticks, which cost one call per market, so we read them for a sample and otherwise take the fill to be at the market's creation hour and its last price. On 398 sampled regular-season markets, 61% fill once, the first fill is in the creation hour about half the time (a median of two hours later otherwise), and the last price is the last fill 92% of the time. For a market created before kickoff this is a fair pre-game price, except when its last trade came during the games, after some legs were decided; about 2% of markets look like that, and they matter for one of the results below.

3.4.3 A worked example: week 10 of 2025

Table 14 shows three games with the four probabilities and the result. Take "Broncos win and Dolphins win". The book gives $0.800 \times 0.213 = 0.171$; our raw model gives $0.792 \times 0.327 = 0.259$; Elo gives $0.852 \times 0.249 = 0.212$. Both teams won, so this parlay hit. "Broncos win and Bills win" was priced $0.800 \times 0.787 = 0.629$ by the book and 0.533 by the model; Miami upset Buffalo, so it missed. Three games give three pairs and four side choices each, twelve parlays in all.

Game	Moneylines	Book	Model	Elo	Result
Raiders at Broncos	DEN −500 / LV +380	DEN 80.0%	79.2%	85.2%	Broncos 10–7
Bills at Dolphins	MIA +350 / BUF −455	MIA 21.3%	32.7%	24.9%	Dolphins 30–13
Rams at 49ers	SF +210 / LA −258	SF 30.9%	39.9%	42.3%	Rams 42–26

Table 14: Home-team win probability from each source, week 10 of 2025. Our model gives the underdog more chance than the book in both games with a clear favourite.

Two ways to score a probability. The *Brier score* is the average of (probability − outcome)^2, with the outcome 1 if the parlay hit and 0 if not. For "Broncos and Dolphins" the book's penalty is $(1 - 0.171)^2 = 0.687$ and the model's $(1 - 0.259)^2 = 0.549$; for "Broncos and Bills" they are $0.629^2 = 0.396$ and $0.533^2 = 0.284$. Lower is better; always saying 50% on a coin flip scores 0.25. The *slope* asks a different question: when a source says 60%, does it happen 60% of the time?

Group parlays by the probability they were given and plot the share that hit against the average probability; a slope of 1 means the source means what it says, above 1 means it is too cautious (its 60% calls hit more than 60%), below 1 means it is over-confident. On the twelve example parlays the Brier scores are 0.207 (book), 0.179 (model) and 0.207 (Elo): the model looks better in this one week because two underdogs won, which is exactly what one week cannot tell you.

3.4.4 Results over 2018–2025

Two-leg parlays (60,384)	Brier	Slope	Both favourites: hit − price	Both underdogs
Closing line, de-vigged	0.166	1.03	−0.2¢ ± 1.7	−0.2¢ ± 0.8
Our model, raw	0.173	1.16	+7.1¢ ± 1.5	−4.3¢ ± 0.7
Our model, recalibrated	0.173	0.86	+3.7¢ ± 1.8	−2.2¢ ± 0.8
Elo	0.173	0.90	+4.4¢ ± 1.6	−2.6¢ ± 0.7

Table 15: Parlay probabilities as the product of each source's single-game probabilities, scored against what happened. Standard errors are clustered by season because parlays in one season share games. Always saying the base rate would score 0.1875.

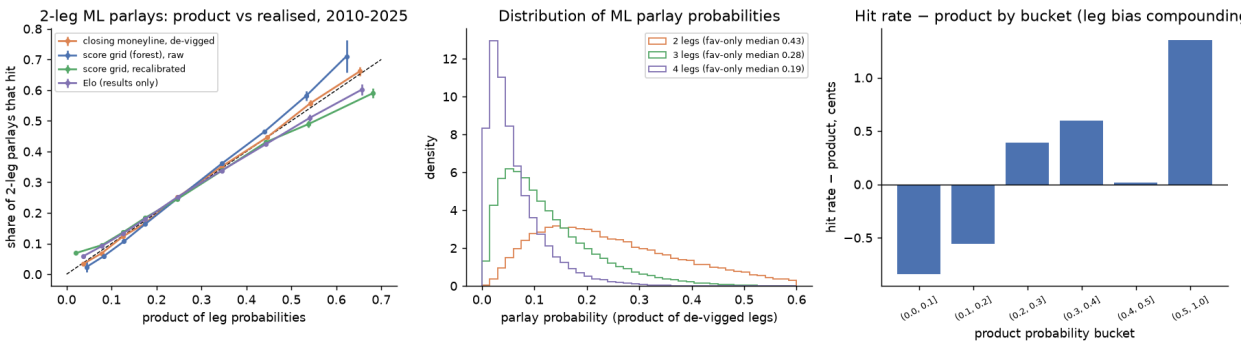


Figure 16: Left: share of two-leg parlays that hit against the product of the leg probabilities, by source. Centre: distribution of parlay probabilities by number of legs; the median favourites-only parlay has probability 0.43 (2 legs), 0.28 (3) and 0.18 (4). Right: hit rate minus the book's product, by probability bucket.

Three things come out.

1. **Multiplying the book's probabilities gives a fair parlay price.** The closing line's product is the most accurate and its slope is 1: a parlay it prices at 40% hits about 40% of the time, for favourites and underdogs alike.
2. **The games are independent.** On the forest's out-of-sample residuals over 2010–2025, the correlation between games on the same date is −0.005 (14,426 pairs, standard error 0.008) and between games in the same week −0.005; for totals −0.017 and −0.008. Two home teams in the same week both win 29.5% of the time against 30.0% from the product. Multiplication is the right rule across games.
3. **Our own probabilities are a little less accurate and biased in a known direction.** All three of our sources score about 0.173 against the book's 0.166. More important is how they are wrong: our model gives underdogs too much chance, so a parlay of two favourites that hits 44% of the time is priced at 37% by the raw model, and the gap doubles with each leg added.

Recalibration halves the gap but does not close it, and Elo, which uses wins alone, behaves the same way. The fix is in the model, not in a correction after the fact: its distribution of margins is too wide for favourites, the same thing the fixed-threshold calibration of Section 3.3 showed with a slope of 1.2.

3.4.5 Same-game legs

Two legs on the same game are not independent, and there the score grid is the model: the parlay's probability is the grid's mass on the intersection of the two legs' regions. Priced on the 2015–2025 grid, the premium over the product of marginals is under half a cent for winner-and-total or spread-and-total combinations (the margin and total of a game are nearly uncorrelated) but 10–25 cents when the legs overlap, such as "home wins" with "home covers −3". We leave same-game parlays for a later report.

3.4.6 How Kalshi prices parlays

For each parlay created before its first kickoff (87% of them) we took the Kalshi price of every leg at the creation hour (the middle of the bid and ask), multiplied them, and compared the product with what the parlay sold for. We did the same with the product of the de-vigged closing moneylines and with the product of our grid's probabilities, the grid being tilted to the Kalshi-implied line at that hour.

Parlay price minus ...	**Mean**	**Median**	**10th–90th pct**	**Within 2¢**
Product of Kalshi leg prices	+1.3¢	+0.1¢	−0.9 to +1.7¢	84%
Product of closing moneylines (de-vigged)	+1.5¢			
Product of our grid's probabilities	+1.7¢	+0.2¢	−0.6 to +3.0¢	81%

Table 16: 694,835 cross-game winner parlays bought before kickoff, weeks 1–8 of 2026. The product of Kalshi's leg prices and the product of the closing moneylines agree with each other to 0.1¢.

Kalshi prices a parlay as the product of its legs. The typical gap is a tenth of a cent. The small positive average has a simple cause: a buyer pays the ask on every leg, and eight half-spreads of a cent each compound to about a cent on the parlay (a regression of the gap on the product and the number of legs gives an intercept of 2.9¢ and −0.2¢ per extra leg, longer parlays being cheaper). The wide tails come from the in-play last prices mentioned above. Our grid's product sits 0.4¢ below the market's, because the model gives underdogs more chance than the market does, which lowers the price of the favourite-heavy parlays people buy.

3.4.7 How the Kalshi parlays settled

Legs	Markets	Price paid	Product of leg prices	Our grid	Share that hit
2	6,438	30.0%	26.1%	25.9%	24.9%
3	23,144	19.0%	14.7%	14.0%	13.4%
4	55,058	12.1%	8.9%	8.2%	7.5%
5–6	184,500	6.7%	4.7%	4.2%	3.3%
7–8	130,565	3.5%	2.4%	2.0%	1.1%
9–12	204,857	1.1%	0.7%	0.5%	0.14%
13 or more	95,429	0.24%	0.09%	0.06%	0.00%
All	699,991	4.6%	3.4%	3.0%	2.4%

Table 17: Winner parlays bought before kickoff and settled, by number of legs: the average price paid, what the product of the leg prices and our grid said, and the share that settled yes. Over all parlays the price exceeds the hit rate by 2.3¢ (95% interval 1.8 to 2.7), the product of leg prices by 1.0¢ (0.5 to 1.6) and the grid by 0.6¢ (0.2 to 1.1); the intervals are from a bootstrap over game dates, since parlays in one week share games.

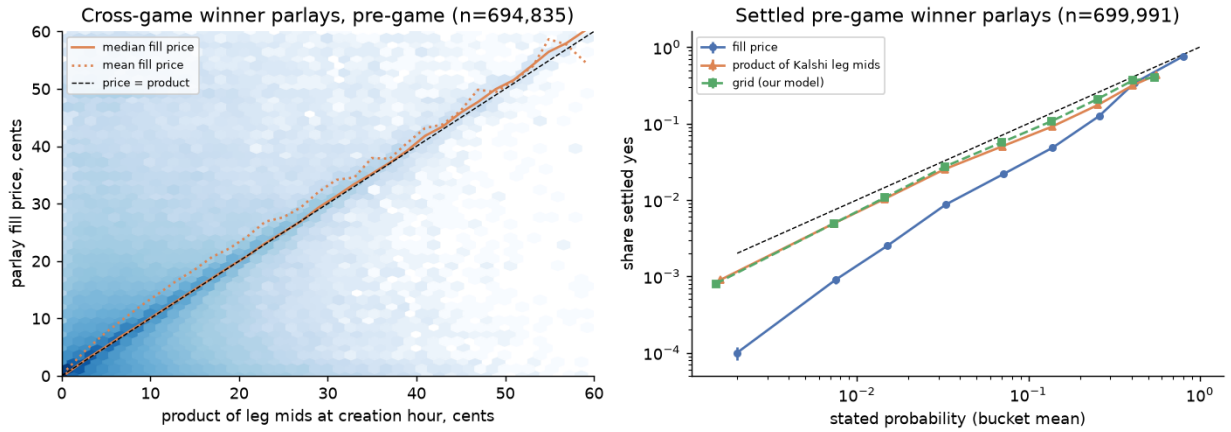


Figure 17: Left: parlay price against the product of the Kalshi leg prices at creation, with the median and mean price in each 2-cent bin; the mass sits on the diagonal. Right: share of parlays that hit against what each source said, by bucket, on log scales; the product of leg prices and the grid run parallel to the diagonal and slightly below it, the price sits further below until the buckets above 35¢, which are filled by in-play last trades.

Three things come out of Table 17.

1. **Buyers pay about twice the realised frequency.** Parlays bought at an average of 4.6¢ hit 2.4% of the time. About a cent of the gap is the spread paid on the legs; the rest is that the legs themselves hit less often than priced this season.
2. **Two-leg parlays settled as priced; long ones did not.** For two legs, the product of leg prices and the grid are within noise of the hit rate. The shortfall grows with the leg count, to a factor of five at 9–12 legs. The reason is compounding of a small error per leg: in weeks 1–7 of 2026 the favourite, priced at 64.7% on average, won 63.2% of games (106 games), and favourites priced above 70% won 67% against 77% priced (30 games), whereas over 2018–2025

favourites won exactly as often as priced (66.6% against 66.5%, 2,119 games). A two-point shortfall per leg is one season's ordinary noise, but raised to the eighth power it is a 20–30% shortfall on the parlay, and the hit rate fell short of the grid in six of the seven weeks. Seven weeks cannot say whether 2026 is unusual or whether parlay buyers pick overpriced legs; the eight-season history above says the product is calibrated across seasons, which favours the first reading.

3. **Our grid was closer to the hit rate than the market this season.** This is the opposite of the history in Table 15, and for the same reason: the model gives favourites less chance than the market, and in these weeks favourites under-delivered. On the 186 markets whose fill time is known from the candles, so that the price is certainly pre-game, the grid's Brier score is 0.0150 against 0.0155 for the price paid. On the full set the price scores better (0.0145 against 0.0217), but that is the in-play last prices at work, not foresight.

3.4.8 Next steps

Narrow the model's margin distribution for favourites, since the direction of its error is now known from both the history and the market; extend the parlay tests to spread and total legs, where the same-game correlation premium matters; and read the candles for a larger sample so that fill times are exact.

4 What we have learned, and next steps

4.1 What we have learned

1. For a binary contract, price, probability and the profit-and-loss distribution are the same thing; for a portfolio on one game, a joint distribution over the final score is the risk model. VaR and expected shortfall are then exact sums, with no covariance matrix or simulation.
2. The NFL score distribution is discrete in ways that matter for pricing: 3-point margins are 2.9 times as likely as a normal model says, and margin and total always share parity. An empirical score grid, repositioned by exponential tilting, prices a game's whole ladder of contracts to within a cent of the Kalshi order book once the dispersion is calibrated.
3. Point accuracy is capped near the Vegas closing line. Our two-output random forest, a linear model, Elo, a public Elo variant and every stack of them sit 0.1 to 0.4 points behind the line, as the literature has found since 1980. The forest and the linear model tie, so with 7,000 games the information is in the features, and the features are mostly ratings.
4. The forest's two score predictions are calibrated, and their residuals are uncorrelated with standard deviations near 9.5 points that rise with the level of scoring. Those are the parameters needed to turn two predicted scores into a probability for any outcome.
5. The resulting cover and over probabilities are calibrated for outcomes the market has not priced but carry no information about which side of the Vegas number will hit. The engine's value is in pricing the whole distribution consistently, not in picking sides against the market.
6. Leakage is the dominant risk in this kind of work and is easy to introduce from a well-maintained public dataset. A perturbation test and a sanity check against the literature's error levels are both needed.

4.2 Next steps

1. **Probabilistic head.** A model that predicts $(\mu_h, \mu_a, \sigma_h, \sigma_a, \rho)$ per game, trained on the bivariate negative log-likelihood, and used to position and scale the empirical grid. The toy example shows a dispersion parameter removes the remaining tail error.
2. **Remaining suggested methods** on the same feature table: LARS for a sparse linear model, gradient boosting, PCA and UMAP as exploration, and a cascade-correlation network as a final comparison.
3. **Offence and defence ratings** in the Glickman and Stern [8] state-space form, so each predicted score has its own strength input rather than a single Elo; recency weighting for the scoring environment; a stadium lookup for travel distance.
4. **Risk layer.** Represent a book of Kalshi positions as a payoff vector on the score grid; compute ES, worst cell and probability of loss exactly; measure in-play jump risk from the trade tape, where hourly price changes are 17 cents against 1 cent pre-game; add the quoted-spread curve (which scales with $p(1-p)$, not with strike distance) as the liquidation cost.
5. **Live record.** Update weekly through the 2026 season: new games, new Kalshi prices, re-run the walk-forward, accumulate out-of-sample forecasts against the market.

References

[1] Philippe Artzner, Freddy Delbaen, Jean-Marc Eber, and David Heath. Coherent measures of risk. *Mathematical Finance*, 9(3):203–228, 1999.

[2] Rose D. Baker and Ian G. McHale. Forecasting exact scores in National Football League games. *International Journal of Forecasting*, 29(1):122–130, 2013.

[3] Bryan L. Boulier and Herman O. Stekler. Predicting the outcomes of National Football League games. *International Journal of Forecasting*, 19(2):257–270, 2003.

[4] Peter Divos, Sebastian Del Bano Rollin, Zsolt Bihari, and Tomaso Aste. Risk-neutral pricing and hedging of in-play football bets. *Applied Mathematical Finance*, 25(4):315–335, 2018.

[5] Mark J. Dixon and Stuart G. Coles. Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C*, 46(2):265–280, 1997.

[6] FiveThirtyEight. NFL Elo ratings: data and methodology. `https://github.com/fivethirtyeight/data/tree/master/nfl-elo`, 2023.

[7] René Manassé Galekwa et al. A systematic review of machine learning in sports betting: techniques, challenges, and future directions. arXiv:2410.21484, 2024.

[8] Mark E. Glickman and Hal S. Stern. A state-space model for National Football League scores. *Journal of the American Statistical Association*, 93(441):25–35, 1998.

[9] Mark E. Glickman and Hal S. Stern. Estimating team strength in the NFL. In *Handbook of Statistical Methods and Analyses in Sports*, pages 113–136. CRC Press, 2017.

[10] Philip K. Gray and Stephen F. Gray. Testing market efficiency: evidence from the NFL sports betting market. *The Journal of Finance*, 52(4):1725–1737, 1997.

[11] Robby Greer. nfelo: an Elo-based NFL prediction model. `https://github.com/greerreNFL/nfelo`, 2026.

[12] David Harville. Predictions for National Football League games via linear-model methodology. *Journal of the American Statistical Association*, 75(371):516–524, 1980.

[13] Steven D. Levitt. Why are gambling markets organised so differently from financial markets? *The Economic Journal*, 114(495):223–246, 2004.

[14] Michael J. Lopez, Gregory J. Matthews, and Benjamin S. Baumer. How often does the best team win? A unified approach to understanding randomness in North American sport. *The Annals of Applied Statistics*, 12(4):2483–2516, 2018.

[15] nflverse. NFL play-by-play, schedule, roster and injury data. `https://nflverse.nflverse.com`, 2026.

[16] Nicholas G. Polson and Hal S. Stern. The implied volatility of a sports game. *Journal of Quantitative Analysis in Sports*, 11(3):145–153, 2015.

[17] Hyun Song Shin. Measuring the incidence of insider trading in a market for state-contingent claims. *The Economic Journal*, 103(420):1141–1153, 1993.

[18] ChiUng Song, Bryan L. Boulier, and Herman O. Stekler. The comparative accuracy of judgmental and model forecasts of American football games. *International Journal of Forecasting*, 23(3):405–413, 2007.

[19] Hal S. Stern. On the probability of winning a football game. *The American Statistician*, 45(3): 179–183, 1991.

[20] Hal S. Stern. A Brownian motion model for the progress of sports scores. *Journal of the American Statistical Association*, 89(427):1128–1134, 1994.

[21] Justin Wolfers and Eric Zitzewitz. Interpreting prediction market prices as probabilities. Technical Report Working Paper 12200, National Bureau of Economic Research, 2006.

[22] Ronald Yurko, Samuel Ventura, and Maksim Horowitz. nflWAR: a reproducible method for offensive player evaluation in football. *Journal of Quantitative Analysis in Sports*, 15(3):163–183, 2019.

[23] Richard A. Zuber, John M. Gandar, and Benny D. Bowers. Beating the spread: testing the efficiency of the gambling market for National Football League games. *Journal of Political Economy*, 93(4):800–806, 1985.